



Group ID : 25J-26-421

IT4010 - Research Project

(Comprehensive Design and Analysis Project)

Project Charter Document

B.Sc. (Hons) in Information Technology specializing in Interactive Media

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

1. Project Title

Evaluating Multimodal Multiplayer Games For Cognitive Social Skills Development In Dyslexic Children.

2. Project Background

Dyslexia is a common learning difficulty that affects reading and writing skills, particularly in early education. Traditional teaching methods and existing educational applications often rely on reactive feedback mechanisms, which provide corrections only after errors occur. This approach can lead to frustration, reduced confidence, and slower learning progress among dyslexic children.

With the advancement of mobile technologies, artificial intelligence, and interactive gaming, there is an opportunity to create a more engaging and effective learning environment. This project aims to address these limitations by developing an integrated system that combines multiplayer interaction, adaptive personalization, reward-based motivation, and anticipatory feedback.

3. Project Purpose / Justification

The purpose of this project is to develop an innovative educational platform that enhances letter tracing skills in dyslexic children through proactive and engaging learning mechanisms.

The system introduces:

- **Anticipatory feedback** to prevent errors before they occur
- **Multiplayer interaction** to encourage collaboration
- **Adaptive learning** to personalize difficulty levels
- **Reward-based Motivation** to increase motivation

This project is justified as it contributes to improving educational accessibility and learning outcomes for children with learning difficulties.

4. Project Objectives

Primary Objective:

To design and develop a mobile-based interactive learning system that improves letter tracing accuracy for dyslexic children using anticipatory multimodal feedback.

Specific Objectives:

- Develop a mobile application for letter tracing
- Implement an Anticipatory Multimodal Feedback System (AMFS)
- Design a multiplayer game environment for collaborative learning
- Integrate adaptive personalization based on user performance
- Implement a reward-based motivation system
- Evaluate system effectiveness using performance metrics

5. Project Scope

In Scope:

- Mobile application development (Flutter-based)
- Letter tracing functionality
- Real-time telemetry data collection
- AMFS implementation (visual, audio, haptic feedback)
- Multiplayer interaction system
- Adaptive difficulty system
- Reward and achievement system
- UI/UX design for dyslexia-friendly interfaces

Out of Scope:

- Large-scale deployment to schools
- Hardware-based assistive devices
- Full-scale AI model training using large datasets
- Clinical validation with medical professionals

6. Deliverables

- Functional mobile application
- Anticipatory Multimodal Feedback System
- Multiplayer game module
- Adaptive personalization module
- Reward system implementation
- UI/UX design mockups and prototypes
- Feasibility report
- Thesis (individual + group)
- Commercialization video with UI/UX demo

7. Project Stakeholders

Stakeholder	Role
Students (Development Team)	Design & development
Supervisor	Guidance & evaluation
Co-Supervisor	Technical review
SLIIT	Academic institution
End Users (Children)	Primary users
Parents/Teachers	Secondary stakeholders

8. Project Team & Responsibilities

Member	Responsibility
IT22597656	AMFS implementation
IT22610034	Multiplayer game development
IT22610034	Adaptive personalization system
IT22599704	Reward-based motivation system

9. High-Level Requirements

- Real-time tracing input processing
- Accurate risk assessment algorithm
- Multimodal feedback delivery (visual, audio, haptic)
- Smooth UI/UX with dyslexia-friendly design
- Multiplayer synchronization
- Performance tracking and evaluation

10. Project Timeline (High-Level)

Phase	Duration
Requirement Analysis	Aug 2025
System Design	Sep 2025
Development Phase 1	Oct – Nov 2025
Development Phase 2	Dec 2025 – Jan 2026
Testing & Refinement	Feb 2026
Final Integration	Mar 2026
Documentation & Submission	Apr 2026

11. Assumptions

- Users have access to Android mobile devices
- Internet connectivity is available for multiplayer features
- Users are within early learning age group
- Flutter framework supports all required functionalities

12. Constraints

- Limited development time
- Limited access to real dyslexic user testing
- Hardware limitations of mobile devices
- Academic deadlines

13. Risks

Risk	Mitigation
Inaccurate feedback system	Improve algorithms and testing
Integration issues	Modular development approach
Performance issues	Optimize code and reduce overhead
Limited user testing	Use simulated testing and feedback

14. Success Criteria

The project will be considered successful if:

- The system accurately detects tracing errors in real time
- Feedback is delivered effectively across multiple modalities
- Users show improved tracing accuracy
- The application runs smoothly on mobile devices
- All components are successfully integrated

15. Approval

Role	Name	Signature	Date
Supervisor	Mr. Didula Chamara		
Co-Supervisor	Mr.Nushkan Nismi		